

A Finite Kneading/Pruning Gate for a Hénon Symbolic Candidate

Anonymous Route-A report

Abstract

We test a source-locked finite kneading candidate for the certified two-branch Hénon interface. Cyclic binary words are admitted only when all lexicographic comparisons are decided by a frozen 32-symbol pair. Through period twelve we enumerate rooted periodic words, invert repetitions exactly, and compute the formal determinant prefix $D(z) = \exp(-\sum_n t_n z^n/n)$. Independent reconstruction and symbolic checks agree. The result is a finite A2 certificate, not an infinite Hénon coding theorem: unresolved comparisons remain excluded, so A1 is open and no analytic Fredholm determinant or arithmetic interpretation is claimed.

1 Frozen candidate

Let $K^- < K^+$ be the frozen lower and upper binary bounds generated from a fixed morphic seed. For a word w of length n , every cyclic suffix is compared with both bounds. A word is accepted only if every comparison is decided within 32 symbols and lies in the interval. This convention makes the finite object deterministic and prevents unresolved tails from being silently promoted.

2 Exact finite ledger

Write t_n for the number of accepted rooted words and p_n for primitive necklaces. The exact repetition identity is

$$t_n = \sum_{d|n} d p_d.$$

The observed rooted counts for $n = 1, \dots, 12$ are

$$0, 2, 6, 14, 30, 62, 112, 238, 456, 902, 1804, 3542.$$

Formal determinant coefficients are generated recursively from

$$n[z^n]D = -\sum_{k=1}^n t_k [z^{n-k}]D, \quad D(0) = 1.$$

Proposition 1. *The producer and an independent implementation agree on every accepted word, primitive count, and determinant coefficient through degree twelve.*

Proof. Both implementations enumerate all binary words directly, apply the same frozen comparison length, and independently check rotation invariance. The divisor identity and the coefficient recurrence then give exact rational cross-checks; a SymPy series calculation agrees coefficient by coefficient. $\square$

object	count
periods checked	12
accepted rooted words at period 12	3542
primitive necklaces at period 12	289
independent checks	5

3 A1/A2 boundary

The finite ledger supports ‘A2_CERTIFIED_PREFIX’. It does not prove that the frozen pair is the infinite kneading invariant of the Hénon repeller, that every accepted word lifts to a geometric orbit, or that the Hofbauer operator is positive recurrent/nuclear. Thus the preregistered verdict is ‘A1_OPEN’, ‘A2_CERTIFIED_PREFIX’, ‘A3_NOT_ADDRESSED’, and ‘A4_FAIL’. No prime table, Riemann-zero table, Euler factor, automorphy statement, or Hilbert–Pólya operator is used.

4 Reproducibility

The JSON evidence, independent checker, SymPy check, replay test, mutation audit, and release manifest are shipped with this paper. A future upgrade must add a source-backed infinite coding theorem and a tail/nuclearity bound for the same transfer object.